

formation about the surfaces, whereas the 3D scanner collects the information about the distance of the various points lying on that surface," explains Singh.

On combining all the data point values, you get a three-dimensional coordinate system for each point, which is good enough to use in any 3D modelling software to generate a well-defined surface. For 3D scanning you need a 3D scanner machine and software to port the scanned data, while for 3D modelling you require the 3D data and 3D modelling software.

In electronic products, quite often the space is limited. Dr Vinay Vaidya, chief technology officer, Engineering & CREST Leader, KPIT Cummins, says, "Fitting the electronics in a given space is a challenge. 3D scanning gives a clear idea about not only the length and width but also the height. Knowing the height of each component gives freedom to the designer in selecting a substitute component meeting the same height constraint. Space available inside the casing is also important from overall cooling perspective."

For most situations, a single scan does not produce a complete model of the subject. Singh says, "Multiple scans, even hundreds, from many different directions are usually required to obtain information about all sides of the subject."

3D model: Reduce cost, develop faster

The cost of product development is increasing with the demand for compact design and multiple functionalities. 3D modelling gives conceptual design, engineering, manufacturing and sales and marketing departments the ability to virtually explore the complete product before it is built.

Ram Bahadur Diwan, a senior designer at Longcheer, explains, "Traditionally, the product development cycle is design→build→test→fix, while in the case of 3D modelling product development steps are design→analyse→test→build. By using 3D modelling to catch design

3D laser scanners & scanning services vendors

- 3Shape
- Artec
- Breukmann
- Capture3D
- Creaform
- Davidscan
- Faro
- GKS
- Handyman
- Konica Minolta
- Mitutoyo
- NextEngine
- Proto3000
- Stienbichler
- Zeiss

problems upfront, manufacturers experience fewer change orders downstream."

3D modelling reduces time, cost and effort for prototype cycles. "Building physical samples or prototypes takes time and money, as a number of prototyping cycles are needed to design a mechanism or enclosure. Today, not more than two-three physical prototype cycles are needed to design a product," shares Vinay Rawat, vice president, product engineering ser-

vices, Wipro Technologies.

Rohit Walke, Jay Instruments, shares, "3D modelling software may vary according to the application. In electronics, it can be used in sheet-metal, rack-mountable enclosure or plastic enclosure design/modelling for electrical equipment like chassis and motherboards."

"For CAD modelling, there are multiple software like ACAD, ProE, CATIA and Solidedge. But for fast and accurate concept design as well as new modeling, a 3D scanning system coupled with redesign software is used," Walke informs.

Talking about prototyping, Singh says, "Any manufacturing process always needs a tolerance limit for production as every machine has its limit of accuracy which makes it impossible to produce a part with zero tolerance. So before starting mass production of any product, we check it while making a prototype. This includes inputs in the form of man, machine, material and cost."

As the cumulative cost of all these variables is very high, "We use 3D modelling technique to visualise a product and to check the position of various features in a product. Also,



Handheld 3D laser scanner